Is it chance?

Calibrating and tuning the group identification

Matteo Di Sante | xc-lab meeting, 23 October 2026

Method note, sections 7–9: `xc-lab/ideas/group-identification`

The question

Two pilots who never interact still meet.
How long must a contact last before it is not chance?

They take off from the same site, follow the same ridges and valleys, climb in the same house thermals. So:

- build **chance** from the data: pairs that fly the same routes without synchrony;
- keep the contacts that are long enough for chance to be **one in ten**;
- check how the result moves with each **threshold**.

Run: 162 485 cleaned flights of 6 209 dates of the FFVL archive; profiles on 300 random dates.

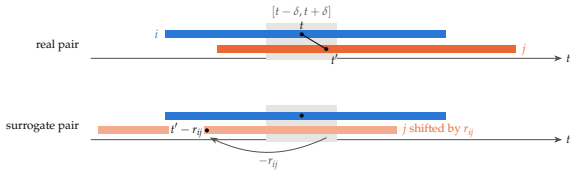
Part 1: what chance means

The null model and the minimum duration τ^*

The null model: shift one flight in time

Surrogate pair. Two flights of the same date, takeoffs less than $R_s = 10$ km apart; the copy of j takes off ϑ after i, with ϑ drawn from the takeoff delays of the day:

$$\mathbf{x}_j^s(t) = \mathbf{x}_j(t + r_{ij}), \qquad r_{ij} = (k_j^0 - k_i^0) - \vartheta, \qquad |r_{ij}| \geq 2\delta + \Delta.$$

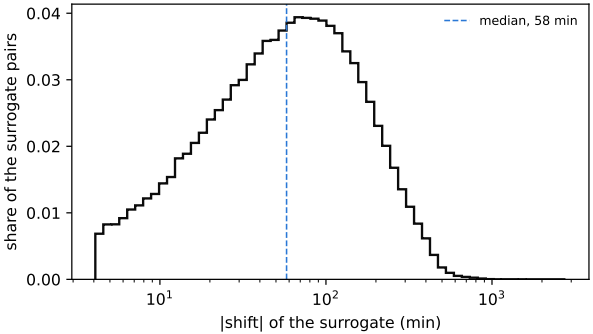


Keeps: route, terrain, date, takeoff delays of the day.

Destroys: every correspondence within δ, since $|r_{ij}| > 2\delta$.

Up to 200 pairs per date. Idea from Spiegel et al. (2016).

The shifts are about an hour



248 741 surrogate pairs of the chosen run, whole archive.

Median shift **58 min**; 90% between 8 min and 4.4 h.

The copy flies in the weather of another hour: other thermals, another cloud base.

So “real” includes pilots who share a thermal at the same time without following each other.

From the surrogates to τ^*

False discovery rate. On each date d: H_d the exposure of the real pairs (time both fly after departure), $\rho_{0,d}(\tau)$ the surrogate intervals lasting at least τ per second of surrogate exposure. With $N(\tau)$ the real intervals lasting at least τ:

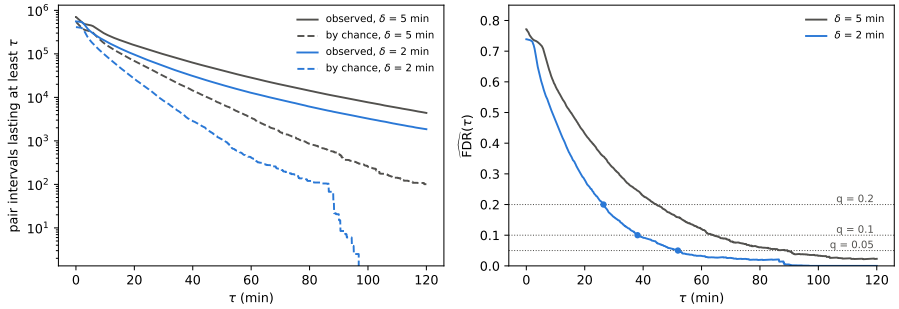
$$C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d, \qquad \widehat{\mathrm{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\{\tau : \widehat{\mathrm{FDR}}(\tau) \le q\}.$$

$C(\tau)$: the intervals lasting at least τ that chance alone would give the real pairs.

τ^*: the shortest duration at which chance is at most a share q of what we observe.

The estimate of SAM (Tusher et al. 2001), with time shifts in place of permutations; rates taken date by date because the surrogates are a sample of each date.

Most short contacts are chance



Left: real intervals lasting at least τ (solid) and chance (dashed), $\delta = 5$ min (grey) and 2 min (blue). Right: their ratio; dots: τ^* for $q = 0.05, 0.1, 0.2$.

Chance is 74% of all pair intervals, 47% of those lasting 10 min, 16% of those lasting 30 min. With $q = 0.1$: $\tau^* = 38.1$ min (37.0–41.1 over 200 resamplings of the dates).

How much the shift matters

Surrogates	pairs	$\widehat{\mathrm{FDR}}$(10 min)	$\widehat{\mathrm{FDR}}$(30 min)	τ^* (min)	5–95%
all shifts	241 824	0.48	0.16	38.2	37.0–41.5
$\lvert r\rvert < 30$ min	69 444	0.73	0.31	53.3	49.7–72.0
$\lvert r\rvert \geq 60$ min	119 687	0.26	0.05	23.0	21.1–24.4

$q = 0.1$, $\delta = 2$ min, on the 2 737 dates with surrogates in each class.

Short shifts keep the weather of the hour, but also bring within δ the pilots who follow each other a few minutes apart. The two effects cannot be separated.

The length of the shift moves τ^* by more than a factor two. We keep the shifts that follow the takeoff delays of the day.

Part 2: the parameters on the data

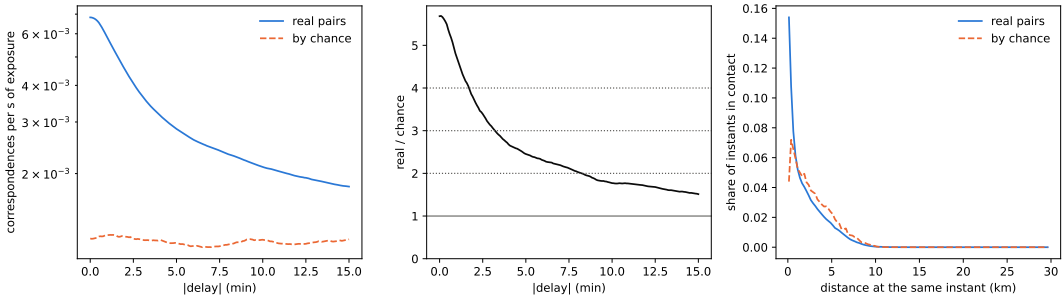
Measured, chosen, or declared as definitions

Three kinds of parameters

Parameter	Value	How it was set
δ, largest delay	2 min	measured on the delay profile
q, accepted false discovery rate	0.1	trade-off between groups found and chance
τ^*, shortest contact	38.1 min	computed from q and the null model
ε_h, ε_z	500 m, 300 m	definitions: profiles without an edge
g, longest gap joined	120 s	definition: gaps of real and chance contacts alike
$\theta_{\max}$	45°	no effect on the result
Δ, $h_{\max}$, R_{dep}, $v_{\max}$, R_s, $r_{\min}$	fixed	physical bounds and exactness

Tolerances say when two positions match; τ^* how long a match must last; the rest makes the computation exact or fast.

δ: the delay profile



300 dates, window of 15 min, 58 910 h of real exposure. Middle: $R(l)$, real over chance correspondences per second of exposure at delay l.

Chance is flat. Real pairs peak at a few seconds, 5.6 times chance; R falls below 4 at 105 s, 3 at 195 s, 2 at 500 s. Fast part: flying together. Slow tail, still 1.5 at 15 min: same thermals at about the same time.

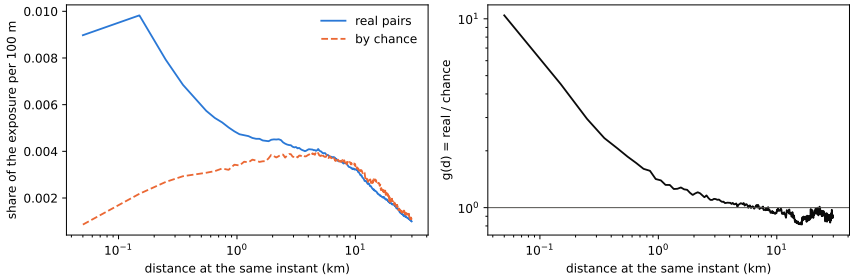
δ: long contacts have short delays

Duration of the interval	intervals	median $\lvert\ell_{\text{med}}\rvert$	within 2 min	at the edge
under 10 min	436 575	210 s	32%	29%
10–30 min	246 587	160 s	39%	12%
30–90 min	102 357	80 s	64%	2.1%
90 min or more	10 734	25 s	90%	6 intervals

ℓ_{med}: median delay of each pair interval, search instants left out. Run with $\delta = 5$ min, whole archive; at the edge: $\lvert\ell_{\text{med}}\rvert \geq 285$ s.

We take $\delta = 2$ min: within it real correspondences are at least 3.7 times chance, and it holds nine in ten of the contacts of 90 min or more.

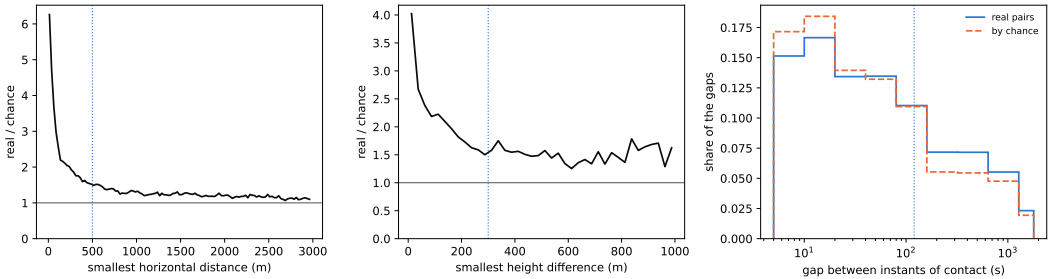
Distance at the same instant



All the time both fly, 210 dates. Right: $g(d)$, real over chance.

Real pairs are 10.4 times chance within 100 m and below 1 only from 5.7 km: the excess goes beyond sight (1.5 km), so it may hold the weather of the hour too. In contact, δ bounds the distance of a pair at $\varepsilon_h + v_{\max}\delta \approx 3.8$ km.

Tolerances: no edge



210 dates, $\delta = 2$ min. Left, middle: real over chance by the smallest horizontal distance and height difference to the other path; dotted: values in use. Right: gap lengths.

Horizontal ratio 6.3 within 25 m, below 1.5 from 500 m; vertical 4.0 within 25 m, near 1.5 from 275 m. Gaps have the same lengths in real and chance contacts. So ε_h, ε_z and g are definitions.

One parameter at a time: no plateau

Setting	τ^* (min)	f_G	episodes
starting values	126.1	0.034	98
$\varepsilon_h = 250$ m	75.1	0.058	253
$\varepsilon_h = 1000$ m	126.8	0.065	162
$\varepsilon_z = 150$ m	126.1	0.020	61
$\varepsilon_z = 600$ m	126.1	0.044	122
$\delta = 2$ min	42.6	0.106	600
$\delta = 10$ min	128.4	0.062	153
$\theta_{\max} = 30°$	126.1	0.034	97
$\theta_{\max} = 90°$	126.1	0.034	98
$g = 60$ s	75.8	0.087	332
$g = 300$ s	126.1	0.055	141

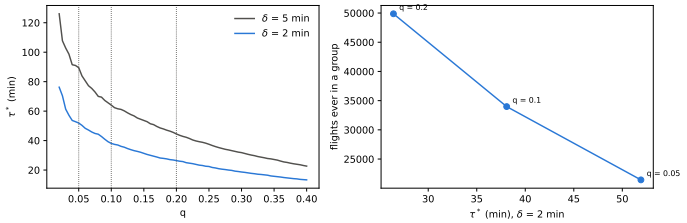
300 random dates (7 051 flights), $\delta = 5$ min unless stated, $q = 0.05$, τ^* recalibrated each time. Not repeated with $\delta = 2$ min.

f_G: fraction of the flight time after departure spent in a group.

$\theta_{\max}$ changes nothing: few contacts in glide.

The others move f_G by up to a factor three, mostly through τ^*.

q: how much chance we accept



δ	q	τ^* (min)	5–95%	para in a group	hang in a group	episodes
5 min	0.05	89.5	79.5–91.5	8.9%	2.1%	6 090
2 min	0.05	51.9	49.1–53.3	13.7%	3.8%	10 553
2 min	0.1	38.1	37.0–41.1	21.7%	6.9%	17 169
2 min	0.2	26.4	25.2–27.2	31.7%	13.9%	26 504

We take $q = 0.1$: at most one contact in ten is chance. With 0.2, chance would make one group in five.

Limits

- Chance is what the null model says: its shifts of about an hour also remove the weather of the hour, so “real” includes pilots in the same thermal at the same time, which is not proof of interaction; τ^* is 23 min with shifts over an hour, 53 min under 30 min.
- The profiles of δ and of the tolerances are on one sample of 300 dates.
- The plateau check is on 300 dates, with the starting $\delta = 5$ min.
- The data are the cleaned tracks of the old repository; the phases that the heading condition needs exist on 1 Hz flights only.

In short

- **Chance**: a pair of the same day with one flight shifted by about an hour. It keeps route, terrain and date, and destroys synchrony.
- **Duration**: most short contacts are chance; at $q = 0.1$ a contact must last $\tau^* = 38.1$ min.
- **Delay**: real pairs match within a few tens of seconds; $\delta = 2$ min holds nine in ten of the long contacts.
- **Tolerances**: no edge and no plateau: ε_h, ε_z and g are definitions, and they move the result by up to a factor three, mostly through τ^*.

21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes.